1) a)

$4\frac{\mathrm{d}^2p}{\mathrm{d}t^2} + 4\frac{\mathrm{d}p}{\mathrm{d}t} + p = 0$

Auxiliary Equation:

$4m^2 + 4m + 1 = 0$

$\Rightarrow (2m+1)^2 = 0$

$\Rightarrow m = -\frac{1}{2}$

General Solution:

$p = At\mathrm{e}^{-\frac{1}{2}t} + B\mathrm{e}^{-\frac{1}{2}t}$

1) b)

Substitute $p = 10$ and $t = 0$:

$10 = A \times 0 \times \mathrm{e}^0 + B\mathrm{e}^0$

$\Rightarrow B = 10$

$\therefore p = At\mathrm{e}^{-\frac{1}{2}t} + 10\mathrm{e}^{-\frac{1}{2}t}$

$\frac{\mathrm{d}p}{\mathrm{d}t} = -\frac{1}{2}At\mathrm{e}^{-\frac{1}{2}t} + A\mathrm{e}^{-\frac{1}{2}t} - 5\mathrm{e}^{-\frac{1}{2}t}$

Substitute $\frac{\mathrm{d}p}{\mathrm{d}t} = -2$ and $t = 0$:

$-2 = -\frac{1}{2}A \times 0 \times \mathrm{e}^0 + A\mathrm{e}^0 - 5\mathrm{e}^0$

$\Rightarrow -2 = A - 5$

$\Rightarrow A = 3$

$\therefore p = 3t\mathrm{e}^{-\frac{1}{2}t} + 10\mathrm{e}^{-\frac{1}{2}t}$

1) c)

For all $t \geq 0$, $\mathrm{e}^{-\frac{1}{2}t} > 0$

So $3t\mathrm{e}^{-\frac{1}{2}t} + 10\mathrm{e}^{-\frac{1}{2}t} > 0$ for all $t \geq 0$

So the price p never becomes negative, according to the model.

1) d)

$p = 3te^{-\frac{1}{2}t} + 10e^{-\frac{1}{2}t}$

$\frac{dp}{dt} = -\frac{3}{2}te^{-\frac{1}{2}t} + 3e^{-\frac{1}{2}t} - 5e^{-\frac{1}{2}t}$

$= -\frac{3}{2}te^{-\frac{1}{2}t} - 2e^{-\frac{1}{2}t}$

$\frac{d^2p}{dt^2} = \frac{3}{4}te^{-\frac{1}{2}t} - \frac{3}{2}e^{-\frac{1}{2}t} + e^{-\frac{1}{2}t}$

$= \frac{3}{4}te^{-\frac{1}{2}t} - \frac{1}{2}e^{-\frac{1}{2}t}$

The rate of decrease is at its greatest when $\frac{d^2p}{dt^2} = 0$

$\Rightarrow \frac{3}{4}te^{-\frac{1}{2}t} - \frac{1}{2}e^{-\frac{1}{2}t} = 0$

$\Rightarrow 3te^{-\frac{1}{2}t} - 2e^{-\frac{1}{2}t} = 0$

$\Rightarrow (3t-2)e^{-\frac{1}{2}t} = 0$

$\Rightarrow 3t - 2 = 0$

$\Rightarrow t = \frac{2}{3}$

So the rate of decrease is at its greatest when $t = \frac{2}{3}$ days

1) e)

One limitation is that the model assumes the price changes continuously and according to fixed coefficients throughout the time period. It does not consider promotions, competitor prices, stock levels, supplier decisions, customer demand, etc.

2) a)

$\frac{d^2\theta}{dt^2} + 5\frac{d\theta}{dt} + 6\theta = 0$

Auxiliary Equation:

$m^2 + 5m + 6 = 0$

$\Rightarrow (m+2)(m+3) = 0$

$\Rightarrow m = -2$ or $m = -3$

General Solution:

$\theta = Ae^{-2t} + Be^{-3t}$

2) b)

Substitute $\theta = 0.5$ and $t = 0$:

$0.5 = Ae^0 + Be^0$

$\Rightarrow A + B = 0.5$ (1)

$\frac{d\theta}{dt} = -2Ae^{-2t} - 3Be^{-3t}$

Substitute $\frac{d\theta}{dt} = 0$ and $t = 0$:

$0 = -2Ae^0 - 3Be^0$

$\Rightarrow 2A + 3B = 0$ (2)

Solve (1) & (2)

$A = \frac{3}{2}$ and $B = -1$

$\therefore \theta = \frac{3}{2}e^{-2t} - e^{-3t}$

2) c)

$\theta = \frac{3}{2}e^{-2t} - e^{-3t}$

$\frac{d\theta}{dt} = -3e^{-2t} + 3e^{-3t}$

$\frac{d^2\theta}{dt^2} = 6e^{-2t} - 9e^{-3t}$

The angular speed is at its greatest when $\frac{d^2\theta}{dt^2} = 0$

$\Rightarrow 6e^{-2t} - 9e^{-3t} = 0$

$\Rightarrow 6e^{t} - 9 = 0$

$\Rightarrow e^{t} = \frac{3}{2}$

$\Rightarrow t = \ln\frac{3}{2}$

$= 0.405$ (3 SF)

So the angular speed is at its greatest when $t = 0.405$ seconds (3 SF)

2) d)

$\left.\frac{d\theta}{dt}\right|_{t=1} = -3e^{-2\times1} + 3e^{-3\times1}$

$= (-3e^{-2} + 3e^{-3})$ rad s^{-1}

$= -0.257$ rad s^{-1} (3 SF)

2) e)

As $t \to \infty$, $\mathrm{e}^{-2t} \to 0$ and $\mathrm{e}^{-3t} \to 0$

So $\theta(t) \to 0$

This means the model predicts that the door approaches the closed position, which is reasonable. However, a real door will come to rest exactly at the closed position, and stop suddenly, rather than follow a perfectly smooth exponential model for all time.

3) a)

$9\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + 6\frac{\mathrm{d}\theta}{\mathrm{d}t} + 2\theta = 0$

Auxiliary Equation:

$9m^2 + 6m + 2 = 0$

$\Rightarrow m^2 + \frac{2}{3}m + \frac{2}{9} = 0$

$\Rightarrow \left(m + \frac{1}{3}\right)^2 - \frac{1}{9} + \frac{2}{9} = 0$

$\Rightarrow \left(m + \frac{1}{3}\right)^2 = -\frac{1}{9}$

$\Rightarrow m = -\frac{1}{3} \pm \frac{1}{3}\mathrm{i}$

General Solution:

$\theta = \mathrm{e}^{-\frac{1}{3}t}\left(A\cos\frac{1}{3}t + B\sin\frac{1}{3}t\right)$

3) b)

Substitute $\theta = 0$ and $t = 0$:

$0 = \mathrm{e}^0(A\cos 0 + B\sin 0)$

$\Rightarrow A = 0$

$\therefore \theta = B\mathrm{e}^{-\frac{1}{3}t}\sin\frac{1}{3}t$

$\frac{\mathrm{d}\theta}{\mathrm{d}t} = \frac{1}{3}B\mathrm{e}^{-\frac{1}{3}t}\cos\frac{1}{3}t - \frac{1}{3}B\mathrm{e}^{-\frac{1}{3}t}\sin\frac{1}{3}t$

Substitute $\frac{\mathrm{d}\theta}{\mathrm{d}t} = 3$ and $t = 0$:

$3 = \frac{1}{3}B\mathrm{e}^0\cos 0 - \frac{1}{3}B\mathrm{e}^0\sin 0$

$\Rightarrow 3 = \frac{1}{3}B$

$\Rightarrow B = 9$

$\therefore \theta = 9\mathrm{e}^{-\frac{1}{3}t}\sin\frac{1}{3}t$

3) c)

The particular solution is $\theta = 9\mathrm{e}^{-\frac{1}{3}t}\sin\frac{1}{3}t$

This contains the sine term $\sin\frac{1}{3}t$ which changes sign repeatedly as t increases.

Therefore $\theta(t)$ alternates between positive and negative values, so the carriage moves repeatedly from one side of equilibrium to the other.

Hence the motion is oscillatory.

Also, because of the factor $\mathrm{e}^{-\frac{1}{3}t}$, the size of the oscillations decreases with time.

3) d)

$\theta = 9\mathrm{e}^{-\frac{1}{3}t}\sin\frac{1}{3}t$

$\Rightarrow \frac{\mathrm{d}\theta}{\mathrm{d}t} = 3\mathrm{e}^{-\frac{1}{3}t}\cos\frac{1}{3}t - 3\mathrm{e}^{-\frac{1}{3}t}\sin\frac{1}{3}t$

The angular displacement is at a maximum when $\frac{\mathrm{d}\theta}{\mathrm{d}t} = 0$

$\Rightarrow 3\mathrm{e}^{-\frac{1}{3}t}\cos\frac{1}{3}t - 3\mathrm{e}^{-\frac{1}{3}t}\sin\frac{1}{3}t = 0$

$\Rightarrow \mathrm{e}^{-\frac{1}{3}t}\cos\frac{1}{3}t - \mathrm{e}^{-\frac{1}{3}t}\sin\frac{1}{3}t = 0$

$\Rightarrow \left(\cos\frac{1}{3}t - \sin\frac{1}{3}t\right)\mathrm{e}^{-\frac{1}{3}t} = 0$

$\Rightarrow \cos\frac{1}{3}t - \sin\frac{1}{3}t = 0$

$\Rightarrow \frac{\sin\frac{1}{3}t}{\cos\frac{1}{3}t} = 1$

$\Rightarrow \tan\frac{1}{3}t = 1$

$\Rightarrow t = 3\arctan 1 = \frac{3\pi}{4}$

$\theta|_{t=\frac{3\pi}{4}} = 9\mathrm{e}^{-\frac{1}{3}\times\frac{3\pi}{4}}\sin\left(\frac{1}{3}\times\frac{3\pi}{4}\right)$

$= 9\mathrm{e}^{-\frac{\pi}{4}}\sin\frac{\pi}{4}$

$= \frac{9\sqrt{2}}{2}\mathrm{e}^{-\frac{\pi}{4}}$ rad

$= 2.90$ rad (3 SF)

3) e)

The carriage comes instantaneously to rest when $\frac{\mathrm{d}\theta}{\mathrm{d}t} = 0$

From part d), $\tan\frac{1}{3}t = 1$ which has a period of $\frac{\pi}{\frac{1}{3}} = 3\pi$

So the 2nd value of $t > 0$ that solves this equation is:

$t = 3\arctan 1 + 3\pi$

$= \frac{15\pi}{4}$ seconds

$= 11.8$ seconds (3 SF)

4) a)

$\frac{\mathrm{d}^2h}{\mathrm{d}t^2} + \frac{\mathrm{d}h}{\mathrm{d}t} + \frac{2}{9}h = 0$

Auxiliary Equation:

$m^2 + m + \frac{2}{9} = 0$

$\Rightarrow \left(m + \frac{1}{2}\right)^2 - \frac{1}{4} + \frac{2}{9} = 0$

$\Rightarrow \left(m + \frac{1}{2}\right)^2 = \frac{1}{36}$

$\Rightarrow m = -\frac{1}{2} \pm \frac{1}{6}$

$\Rightarrow m = -\frac{1}{3}$ or $m = -\frac{2}{3}$

General Solution:

$h = A\mathrm{e}^{-\frac{1}{3}t} + B\mathrm{e}^{-\frac{2}{3}t}$

4) b)

Substitute $h = 0.3$ and $t = 0$:

$0.3 = A\mathrm{e}^0 + B\mathrm{e}^0$

$\Rightarrow A + B = \frac{3}{10}$ (1)

$\frac{\mathrm{d}h}{\mathrm{d}t} = -\frac{1}{3}A\mathrm{e}^{-\frac{1}{3}t} - \frac{2}{3}B\mathrm{e}^{-\frac{2}{3}t}$

Substitute $\frac{\mathrm{d}h}{\mathrm{d}t} = -0.3$ and $t = 0$:

$-0.3 = -\frac{1}{3}A\mathrm{e}^0 - \frac{2}{3}B\mathrm{e}^0$

$\Rightarrow -\frac{1}{3}A - \frac{2}{3}B = -\frac{3}{10}$ (2)

Solve (1) & (2)

$A = -\frac{3}{10}$ and $B = \frac{3}{5}$

$\therefore h = -\frac{3}{10}\mathrm{e}^{-\frac{1}{3}t} + \frac{3}{5}\mathrm{e}^{-\frac{2}{3}t}$

4) c)

The water level is at equilibrium when $h = 0$

$-\frac{3}{10}\mathrm{e}^{-\frac{1}{3}t} + \frac{3}{5}\mathrm{e}^{-\frac{2}{3}t} = 0$

$\Rightarrow -\mathrm{e}^{-\frac{1}{3}t} + 2\mathrm{e}^{-\frac{2}{3}t} = 0$

$\Rightarrow -\mathrm{e}^{\frac{1}{3}t} + 2 = 0$

$\Rightarrow \mathrm{e}^{\frac{1}{3}t} = 2$

$\Rightarrow \frac{1}{3}t = \ln 2$

$\Rightarrow t = 3\ln 2$ seconds

$= 2.08$ seconds (3 SF)

4) d)

$h = -\frac{3}{10}\mathrm{e}^{-\frac{1}{3}t} + \frac{3}{5}\mathrm{e}^{-\frac{2}{3}t}$

$\frac{\mathrm{d}h}{\mathrm{d}t} = \frac{1}{10}\mathrm{e}^{-\frac{1}{3}t} - \frac{2}{5}\mathrm{e}^{-\frac{2}{3}t}$

The greatest deviation of the water level below equilibrium will occur when $\frac{\mathrm{d}h}{\mathrm{d}t} = 0$

$\frac{1}{10}\mathrm{e}^{-\frac{1}{3}t} - \frac{2}{5}\mathrm{e}^{-\frac{2}{3}t} = 0$

$\Rightarrow \mathrm{e}^{-\frac{1}{3}t} - 4\mathrm{e}^{-\frac{2}{3}t} = 0$

$\Rightarrow \mathrm{e}^{\frac{1}{3}t} - 4 = 0$

$\Rightarrow \mathrm{e}^{\frac{1}{3}t} = 4$

$\Rightarrow \frac{1}{3}t = \ln 4$

$\Rightarrow t = 3\ln 4$

$= 4.16$ (3 SF)

$h|_{t=3\ln 4} = -\frac{3}{10}\mathrm{e}^{-\frac{1}{3}\times 3\ln 4} + \frac{3}{5}\mathrm{e}^{-\frac{2}{3}\times 3\ln 4}$

$= -\frac{3}{10}\mathrm{e}^{-\ln 4} + \frac{3}{5}\mathrm{e}^{-2\ln 4}$

$= -\frac{3}{10}\mathrm{e}^{\ln(4^{-1})} + \frac{3}{5}\mathrm{e}^{\ln 4^{-2}}$

$= -\frac{3}{10}\times\frac{1}{4} + \frac{3}{5}\times\frac{1}{16}$

$= -\frac{3}{80}$ m

So $-\frac{3}{80} = -0.0375$ m

So the greatest deviation of the water level below equilibrium is 0.0375 m, and this occurs when $t = 4.16$ seconds (3 SF).

4) e)

$h = -\frac{3}{10}\mathrm{e}^{-\frac{1}{3}t} + \frac{3}{5}\mathrm{e}^{-\frac{2}{3}t}$

$\frac{\mathrm{d}h}{\mathrm{d}t} = \frac{1}{10}\mathrm{e}^{-\frac{1}{3}t} - \frac{2}{5}\mathrm{e}^{-\frac{2}{3}t}$

$\left.\frac{\mathrm{d}h}{\mathrm{d}t}\right|_{t=3} = \frac{1}{10}\mathrm{e}^{-\frac{1}{3}\times 3} - \frac{2}{5}\mathrm{e}^{-\frac{2}{3}\times 3}$

$= \frac{1}{10}\mathrm{e}^{-1} - \frac{2}{5}\mathrm{e}^{-2}$ ms^{-1}

$= -0.0173$ ms^{-1} (3 SF)

5) a)

$\frac{d^2x}{dt^2} + a\frac{dx}{dt} + bx = 0$

Substitute $x = -0.02$, $\frac{dx}{dt} = 0$ and $\frac{d^2x}{dt^2} = 8$

$8 + a \times 0 + b \times -0.02 = 0$

$\Rightarrow 8 - 0.02b = 0$

$\Rightarrow b = 400$

$\therefore \frac{d^2x}{dt^2} + a\frac{dx}{dt} + 400x = 0$

Auxiliary Equation:

$m^2 + am + 400 = 0$

Repeated Roots $\Rightarrow b^2 - 4ac = 0$

So $a^2 - 4 \times 1 \times 400 = 0$

$\Rightarrow a^2 = 1600$

$\Rightarrow a = 40$

$\therefore \frac{d^2x}{dt^2} + 40\frac{dx}{dt} + 400x = 0$

So $a = 40$ and $b = 400$

5) b)

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 40\frac{\mathrm{d}x}{\mathrm{d}t} + 400x = 0$

Auxiliary Equation:

$m^2 + 40m + 400 = 0$

$\Rightarrow (m + 20)^2 = 0$

$\Rightarrow m = -20$

General Solution:

$x = At\mathrm{e}^{-20t} + B\mathrm{e}^{-20t}$

Substitute $x = -0.02$ and $t = 0$

$-0.02 = A \times 0 \times \mathrm{e}^0 + B\mathrm{e}^0$

$\Rightarrow B = -0.02 = -\frac{1}{50}$

$\therefore x = At\mathrm{e}^{-20t} - \frac{1}{50}\mathrm{e}^{-20t}$

$\frac{\mathrm{d}x}{\mathrm{d}t} = -20At\mathrm{e}^{-20t} + A\mathrm{e}^{-20t} + \frac{2}{5}\mathrm{e}^{-20t}$

Substitute $\frac{\mathrm{d}x}{\mathrm{d}t} = 0$ and $t = 0$

$0 = -20A \times 0 \times \mathrm{e}^0 + A\mathrm{e}^0 + \frac{2}{5}\mathrm{e}^0$

$\Rightarrow 0 = A + \frac{2}{5}$

$\Rightarrow A = -\frac{2}{5}$

$\therefore x = -\frac{2}{5}t\mathrm{e}^{-20t} - \frac{1}{50}\mathrm{e}^{-20t}$

5) c)

As $t \to \infty$, $\mathrm{e}^{-20t} \to 0$ and $t\mathrm{e}^{-t} \to 0$ (as e^{-t} tends to zero faster even though t is increasing)

$\therefore x \to 0$ as $t \to \infty$

The displacement never becomes positive, which means the camera does not pass through equilibrium and does not oscillate.

So the camera approaches its equilibrium position as time increases.

6) a)

$\frac{\mathrm{d}^2v}{\mathrm{d}t^2}+\frac{3}{2}\frac{\mathrm{d}v}{\mathrm{d}t}+\frac{1}{2}v=4$

Homogeneous Equation:

$\frac{\mathrm{d}^2v}{\mathrm{d}t^2}+\frac{3}{2}\frac{\mathrm{d}v}{\mathrm{d}t}+\frac{1}{2}v=0$

Auxiliary Equation:

$m^2+\frac{3}{2}m+\frac{1}{2}=0$

$\Rightarrow \left(m+\frac{3}{4}\right)^2-\frac{9}{16}+\frac{1}{2}=0$

$\Rightarrow \left(m+\frac{3}{4}\right)^2=\frac{1}{16}$

$\Rightarrow m=-\frac{3}{4}\pm\frac{1}{4}$

$\Rightarrow m=-\frac{1}{2}$ or $m=-1$

Complementary Function:

$v=A\mathrm{e}^{-\frac{1}{2}t}+B\mathrm{e}^{-t}$

Try this Particular Integral:

$v=k$

$\frac{\mathrm{d}v}{\mathrm{d}t}=0$

$\frac{\mathrm{d}^2v}{\mathrm{d}t^2}=0$

$\frac{\mathrm{d}^2v}{\mathrm{d}t^2}+\frac{3}{2}\frac{\mathrm{d}v}{\mathrm{d}t}+\frac{1}{2}v=4$

$\Rightarrow (0)+\frac{3}{2}(0)+\frac{1}{2}(k)=4$

$\Rightarrow \frac{1}{2}k=4$

$\Rightarrow k=8$

Particular Integral:

$v=8$

General Solution:

$v=A\mathrm{e}^{-\frac{1}{2}t}+B\mathrm{e}^{-t}+8$

6) b)

$v = Ae^{-\frac{1}{2}t} + Be^{-t} + 8$

Substitute $v = 0$ and $t = 0$

$0 = Ae^{0} + Be^{0} + 8$

$\Rightarrow A + B = -8$ (1)

$\frac{dv}{dt} = -\frac{1}{2}Ae^{-\frac{1}{2}t} - Be^{-t}$

Substitute $\frac{dv}{dt} = 1$ and $t = 0$

$1 = -\frac{1}{2}Ae^{0} - Be^{0}$

$\Rightarrow -\frac{1}{2}A - B = 1$ (2)

Solve (1) & (2)

$A = -14$ and $B = 6$

$\therefore v = -14e^{-\frac{1}{2}t} + 6e^{-t} + 8$

6) c)

$\frac{dv}{dt} = 7e^{-\frac{1}{2}t} - 6e^{-t}$

$\frac{d^2v}{dt^2} = -\frac{7}{2}e^{-\frac{1}{2}t} + 6e^{-t}$

The greatest acceleration occurs when $\frac{d^2v}{dt^2} = 0$

$-\frac{7}{2}e^{-\frac{1}{2}t} + 6e^{-t} = 0$

$\Rightarrow -\frac{7}{2}e^{\frac{1}{2}t} + 6 = 0$

$\Rightarrow e^{\frac{1}{2}t} = \frac{12}{7}$

$\Rightarrow \frac{1}{2}t = \ln\frac{12}{7}$

$\Rightarrow t = 2\ln\frac{12}{7}$ seconds

$= 1.08$ seconds (3 SF)

6) d)

$v = -14e^{-\frac{1}{2}t} + 6e^{-t} + 8$

As $t \to \infty$, $e^{-\frac{1}{2}t} \to 0$ and $e^{-t} \to 0$

So $v \to 8$

Therefore, the limiting speed is 8 ms^{-1}

6) e)

The model predicts the speed approaches a constant limiting value of 8 ms^{-1}.
A real car would not usually continue under exactly the same conditions. In practice, the motion would be affected by factors such as: changes in driver input, changes in road conditions, gear changes, varying resistance, running out of petrol, and the inability to travel in a straight line indefinitely due to terrain.

7) a)

$\frac{\mathrm{d}^2T}{\mathrm{d}t^2} + 3\frac{\mathrm{d}T}{\mathrm{d}t} + 2T = 12 + t$

Homogeneous Equation:

$\frac{\mathrm{d}^2T}{\mathrm{d}t^2} + 3\frac{\mathrm{d}T}{\mathrm{d}t} + 2T = 0$

Auxiliary Equation:

$m^2 + 3m + 2 = 0$

$\Rightarrow (m+1)(m+2) = 0$

$\Rightarrow m = -1$ or $m = -2$

Complementary Function:

$v = A\mathrm{e}^{-t} + B\mathrm{e}^{-2t}$

Try this Particular Integral:

$T = pt + q$

$\frac{\mathrm{d}T}{\mathrm{d}t} = p$

$\frac{\mathrm{d}^2T}{\mathrm{d}t^2} = 0$

$\frac{\mathrm{d}^2T}{\mathrm{d}t^2} + 3\frac{\mathrm{d}T}{\mathrm{d}t} + 2T = 17 + t$

$\Rightarrow (0) + 3(p) + 2(pt + q) = 17 + t$

$\Rightarrow (2p)t + (3p + 2q) = t + 17$

Compare coefficients:

$2p = 1 \Rightarrow p = \frac{1}{2}$

$3 \times \frac{1}{2} + 2q = 17 \Rightarrow q = \frac{31}{4}$

Particular Integral:

$T = \frac{1}{2}t + \frac{31}{4}$

General Solution:

$T = A\mathrm{e}^{-t} + B\mathrm{e}^{-2t} + \frac{1}{2}t + \frac{31}{4}$

7) b)

$T = Ae^{-t} + Be^{-2t} + \frac{1}{2}t + \frac{31}{4}$

Substitute $T = 8$ and $t = 0$

$8 = Ae^{0} + Be^{0} + \frac{1}{2} \times 0 + \frac{31}{4}$

$\Rightarrow 8 = A + B + \frac{31}{4}$

$\Rightarrow A + B = \frac{1}{4}$ (1)

$\frac{dT}{dt} = -Ae^{-t} - 2Be^{-2t} + \frac{1}{2}$

Substitute $\frac{dT}{dt} = 0.25$ and $t = 0$

$0.25 = -Ae^{0} - 2Be^{0} + \frac{1}{2}$

$\Rightarrow 0.25 = -A - 2B + \frac{1}{2}$

$\Rightarrow A + 2B = \frac{1}{4}$ (2)

Solve (1) & (2)

$A = \frac{1}{4}$ and $B = 0$

$\therefore T = \frac{1}{4}e^{-t} + \frac{1}{2}t + \frac{31}{4}$

7) c)

$T = \frac{1}{4}e^{-t} + \frac{1}{2}t + \frac{31}{4}$

$\frac{dT}{dt} = -\frac{1}{4}e^{-t} + \frac{1}{2}$

The temperature is increasing when $\frac{dT}{dt} > 0$

$\Rightarrow -\frac{1}{4}e^{-t} + \frac{1}{2} > 0$

$\Rightarrow -\frac{1}{4} + \frac{1}{2}e^{t} > 0$

$\Rightarrow \frac{1}{2}e^{t} > \frac{1}{4}$

$\Rightarrow e^{t} > \frac{1}{2}$

As $t \geq 0$, $e^{t} \geq 1$, so $e^{t} > \frac{1}{2}$

So the temperature is always increasing.

7) d)

As $t \to \infty$, $\mathrm{e}^{-t} \to 0$

So the exponential term vanishes and we are left with $T \sim \frac{1}{2}t + \frac{31}{4}$

So $T \to \infty$ as $t \to \infty$

7) e)

The model may be reasonable over a short period if the heating output is steadily increasing, but it would not be realistic for a long time. The heater cannot increase output indefinitely, and the room temperature would not continue rising without limit.

8) a)

$\frac{\mathrm{d}^2h}{\mathrm{d}t^2} + \frac{\mathrm{d}h}{\mathrm{d}t} + \frac{1}{4}h = t^2 + 2$

Homogeneous Equation:

$\frac{\mathrm{d}^2h}{\mathrm{d}t^2} + \frac{\mathrm{d}h}{\mathrm{d}t} + \frac{1}{4}h = 0$

Auxiliary Equation:

$m^2 + m + \frac{1}{4} = 0$

$\Rightarrow \left(m + \frac{1}{2}\right)^2 = 0$

$\Rightarrow m = -\frac{1}{2}$

Complementary Function:

$h = At\mathrm{e}^{-\frac{1}{2}t} + B\mathrm{e}^{-\frac{1}{2}t}$

Try this Particular Integral:

$h = pt^2 + qt + r$

$\frac{\mathrm{d}h}{\mathrm{d}t} = 2pt + q$

$\frac{\mathrm{d}^2h}{\mathrm{d}t^2} = 2p$

$\frac{\mathrm{d}^2h}{\mathrm{d}t^2} + \frac{\mathrm{d}h}{\mathrm{d}t} + \frac{1}{4}h = t^2 + 2$

$\Rightarrow (2p) + (2pt + q) + \frac{1}{4}(pt^2 + qt + r) = t^2 + 2$

$\Rightarrow \left(\frac{1}{4}p\right)t^2 + \left(2p + \frac{1}{4}q\right)t + \left(2p + q + \frac{1}{4}r\right) = t^2 + 2$

Compare coefficients:

$\frac{1}{4}p = 1 \Rightarrow p = 4$

$2 \times 4 + \frac{1}{4}q = 0 \Rightarrow q = -32$

$2 \times 4 - 32 + \frac{1}{4}r = 2$

$\Rightarrow r = 104$

Particular Integral:

$h = 4t^2 - 32t + 104$

General Solution:

$h = At\mathrm{e}^{-\frac{1}{2}t} + B\mathrm{e}^{-\frac{1}{2}t} + 4t^2 - 32t + 104$

8) b)

$h = Ate^{-\frac{1}{2}t} + Be^{-\frac{1}{2}t} + 4t^2 - 32t + 104$

Substitute $h = 0$ and $t = 0$

$0 = A \times 0 \times e^0 + Be^0 + 4 \times (0)^2 - 32 \times 0 + 104$

$\Rightarrow 0 = B + 104$

$\Rightarrow B = -104$

$\therefore h = Ate^{-\frac{1}{2}t} - 104e^{-\frac{1}{2}t} + 4t^2 - 32t + 104$

$\frac{dh}{dt} = -\frac{1}{2}Ate^{-\frac{1}{2}t} + Ae^{-\frac{1}{2}t} + 52e^{-\frac{1}{2}t} + 8t - 32$

Substitute $\frac{dh}{dt} = 0$ and $t = 0$

$0 = -\frac{1}{2}A \times 0 \times e^0 + Ae^0 + 52e^0 + 8 \times 0 - 32$

$\Rightarrow 0 = A + 52 - 32$

$\Rightarrow A = -20$

$\therefore h = -20te^{-\frac{1}{2}t} - 104e^{-\frac{1}{2}t} + 4t^2 - 32t + 104$

8) c)

$h = -20te^{-\frac{1}{2}t} - 104e^{-\frac{1}{2}t} + 4t^2 - 32t + 104$

The rate of change of the water level at time t is $\frac{dh}{dt}$

$\frac{dh}{dt} = 10te^{-\frac{1}{2}t} - 20e^{-\frac{1}{2}t} + 52e^{-\frac{1}{2}t} + 8t - 32$

$= 10te^{-\frac{1}{2}t} + 32e^{-\frac{1}{2}t} + 8t - 32$

$\frac{d^2h}{dt^2} = -5te^{-\frac{1}{2}t} + 10e^{-\frac{1}{2}t} - 16e^{-\frac{1}{2}t} + 8$

$= -5te^{-\frac{1}{2}t} - 6e^{-\frac{1}{2}t} + 8$

$h''(0) = \left.\frac{d^2h}{dt^2}\right|_{t=0} = -5 \times 0 \times e^0 - 6e^0 + 8$

$= 2$

At $t = 0$, the rate of increase of the water level is increasing at 2 ms^{-2}

8) d)

$h = -20te^{-\frac{1}{2}t} - 104e^{-\frac{1}{2}t} + 4t^2 - 32t + 104$

As $t \to \infty$, $e^{-\frac{1}{2}t} \to 0$ and $te^{-\frac{1}{2}t} \to 0$ (as $e^{-\frac{1}{2}t}$ tends to zero faster than t increases)

So $h \sim 4t^2 - 32t + 104$

The dominant term is $4t^2$, where $4t^2 \to \infty$ as $t \to \infty$

So the water level increases without bound.

8) e)

The model predicts that the water level increases without bound as time increases. This is not realistic for a real tank as the tank would have a finite capacity, and the pump output could not continue increasing indefinitely. The model may be useful over a short period of time, but it would not remain valid for large values of t.

9) a)

$80\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2}+40\frac{\mathrm{d}\theta}{\mathrm{d}t}+25\theta=136\cos\frac{1}{2}t$

Homogeneous Equation:

$80\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2}+40\frac{\mathrm{d}\theta}{\mathrm{d}t}+25\theta=0$

Auxiliary Equation:

$80m^2+40m+25=0$

$\Rightarrow m^2+\frac{1}{2}m+\frac{5}{16}=0$

$\Rightarrow \left(m+\frac{1}{4}\right)^2-\frac{1}{16}+\frac{5}{16}=0$

$\Rightarrow \left(m+\frac{1}{4}\right)^2=-\frac{1}{4}$

$\Rightarrow m=-\frac{1}{4}\pm\frac{1}{2}\mathrm{i}$

Complementary Function:

$\theta=\mathrm{e}^{-\frac{1}{4}t}\left(A\cos\frac{1}{2}t+B\sin\frac{1}{2}t\right)$

Try this Particular Integral:

$\theta=p\cos\frac{1}{2}t+q\sin\frac{1}{2}t$

$\frac{\mathrm{d}\theta}{\mathrm{d}t}=-\frac{1}{2}p\sin\frac{1}{2}t+\frac{1}{2}q\cos\frac{1}{2}t$

$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2}=-\frac{1}{4}p\cos\frac{1}{2}t-\frac{1}{4}q\sin\frac{1}{2}t$

$80\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2}+40\frac{\mathrm{d}\theta}{\mathrm{d}t}+25\theta=136\cos\frac{1}{2}t$

$\Rightarrow 80\left(-\frac{1}{4}p\cos\frac{1}{2}t-\frac{1}{4}q\sin\frac{1}{2}t\right)+40\left(-\frac{1}{2}p\sin\frac{1}{2}t+\frac{1}{2}q\cos\frac{1}{2}t\right)+25\left(p\cos\frac{1}{2}t+q\sin\frac{1}{2}t\right)=136\cos\frac{1}{2}t$

$\Rightarrow(-20p+20q+25p)\cos\frac{1}{2}t+(-20q-20p+25q)\sin\frac{1}{2}t=136\cos\frac{1}{2}t$

$\Rightarrow(5p+20q)\cos\frac{1}{2}t+(-20p+5q)\sin\frac{1}{2}t=136\cos\frac{1}{2}t$

Compare coefficients:

$5p+20q=136$ (1)

$-20p+5q=0$ (2)

Solve (1) & (2)

$p=\frac{8}{5}$ and $q=\frac{32}{5}$

Particular Integral:

$\theta=\frac{8}{5}\cos\frac{1}{2}t+\frac{32}{5}\sin\frac{1}{2}t$

General Solution:

$\theta = \mathrm{e}^{-\frac{1}{4}t}\left(A\cos\frac{1}{2}t + B\sin\frac{1}{2}t\right) + \frac{8}{5}\cos\frac{1}{2}t + \frac{32}{5}\sin\frac{1}{2}t$

Substitute $\theta = 0.4$ and $t = 0$

$0.4 = \mathrm{e}^{0}(A\cos 0 + B\sin 0) + \frac{8}{5}\cos 0 + \frac{32}{5}\sin 0$

$\Rightarrow \frac{2}{5} = A + \frac{8}{5}$

$\Rightarrow A = -\frac{6}{5}$

$\therefore \theta = \mathrm{e}^{-\frac{1}{4}t}\left(-\frac{6}{5}\cos\frac{1}{2}t + B\sin\frac{1}{2}t\right) + \frac{8}{5}\cos\frac{1}{2}t + \frac{32}{5}\sin\frac{1}{2}t$

$\frac{\mathrm{d}\theta}{\mathrm{d}t} = \mathrm{e}^{-\frac{1}{4}t}\left(\frac{3}{5}\sin\frac{1}{2}t + \frac{1}{2}B\cos\frac{1}{2}t\right) - \frac{1}{4}\mathrm{e}^{-\frac{1}{4}t}\left(-\frac{6}{5}\cos\frac{1}{2}t + B\sin\frac{1}{2}t\right) - \frac{4}{5}\sin\frac{1}{2}t + \frac{16}{5}\cos\frac{1}{2}t$

Substitute $\frac{d\theta}{dt} = -0.5$ and $t = 0$

$-0.5 = \mathrm{e}^{0}\left(\frac{3}{5}\sin 0 + \frac{1}{2}B\cos 0\right) - \frac{1}{4}\mathrm{e}^{0}\left(-\frac{6}{5}\cos 0 + B\sin 0\right) - \frac{4}{5}\sin 0 + \frac{16}{5}\cos 0$

$\Rightarrow -\frac{1}{2} = \frac{1}{2}B + \frac{3}{10} + \frac{16}{5}$

$\Rightarrow B = -8$

$\therefore \theta = \mathrm{e}^{-\frac{1}{4}t}\left(-\frac{6}{5}\cos\frac{1}{2}t - 8\sin\frac{1}{2}t\right) + \frac{8}{5}\cos\frac{1}{2}t + \frac{32}{5}\sin\frac{1}{2}t$

9) b)

As $t \to \infty$, $\mathrm{e}^{-\frac{1}{4}t} \to 0$

So for large values of t, $\theta \sim \frac{8}{5}\cos\frac{1}{2}t + \frac{32}{5}\sin\frac{1}{2}t$

So the rotating platform approaches a steady oscillation with period $\frac{2\pi}{\frac{1}{2}} = 4\pi$

9) c)

The steady-state motion is $\theta_{ss} = \frac{8}{5}\cos\frac{1}{2}t + \frac{32}{5}\sin\frac{1}{2}t$

$\frac{8}{5}\cos\frac{1}{2}t + \frac{32}{5}\sin\frac{1}{2}t \equiv R\cos\left(\frac{1}{2}t - \alpha\right)$

$\equiv R\left(\cos\frac{1}{2}t\cos\alpha + \sin\frac{1}{2}t\sin\alpha\right)$

$\equiv R\cos\alpha\cos\frac{1}{2}t + R\sin\alpha\sin\frac{1}{2}t$

So:

$R\cos\alpha = \frac{8}{5} \Rightarrow \cos\alpha = \frac{8}{5R}$

$R\sin\alpha = \frac{32}{5} \Rightarrow \sin\alpha = \frac{32}{5R}$

Amplitude $= \sqrt{\left(\frac{8}{5}\right)^2 + \left(\frac{32}{5}\right)^2} = \frac{8\sqrt{17}}{5}$

$\frac{\sin\alpha}{\cos\alpha} = \frac{\frac{32}{5R}}{\frac{8}{5R}} \Rightarrow \tan\alpha = 4 \Rightarrow \alpha = \arctan 4 = 1.33$ (3 SF)

$\therefore \frac{8}{5}\cos\frac{1}{2}t + \frac{32}{5}\sin\frac{1}{2}t \equiv \frac{8\sqrt{17}}{5}\cos\left(\frac{1}{2}t - \arctan 4\right)$

9) d)

$R = \frac{8\sqrt{17}}{5} = 6.596969\ldots > 6.5$

So as the steady-state amplitude is greater than 6.5, the platform cannot be regarded as safe.

10) a)

$36\frac{\mathrm{d}^2T}{\mathrm{d}t^2} + 108\frac{\mathrm{d}T}{\mathrm{d}t} + 81T = 216\mathrm{e}^{-\frac{3}{2}t} + 100\mathrm{e}^{\frac{1}{6}t}$

Homogeneous Equation:

$36\frac{\mathrm{d}^2T}{\mathrm{d}t^2} + 108\frac{\mathrm{d}T}{\mathrm{d}t} + 81T = 0$

Auxiliary Equation:

$36m^2 + 108m + 81 = 0$

$\Rightarrow m^2 + 3m + \frac{9}{4} = 0$

$\Rightarrow \left(m + \frac{3}{2}\right)^2 = 0$

$\Rightarrow m = -\frac{3}{2}$

Complementary Function:

$T = At\mathrm{e}^{-\frac{3}{2}t} + B\mathrm{e}^{-\frac{3}{2}t}$

Try this Particular Integral:

$T = pt^2\mathrm{e}^{-\frac{3}{2}t} + q\mathrm{e}^{\frac{1}{6}t}$

$\frac{\mathrm{d}T}{\mathrm{d}t} = -\frac{3}{2}pt^2\mathrm{e}^{-\frac{3}{2}t} + 2pt\mathrm{e}^{-\frac{3}{2}t} + \frac{1}{6}q\mathrm{e}^{\frac{1}{6}t}$

$\frac{\mathrm{d}^2T}{\mathrm{d}t^2} = \frac{9}{4}pt^2\mathrm{e}^{-\frac{3}{2}t} - 3pt\mathrm{e}^{-\frac{3}{2}t} - 3pt\mathrm{e}^{-\frac{3}{2}t} + 2p\mathrm{e}^{-\frac{3}{2}t} + \frac{1}{36}q\mathrm{e}^{\frac{1}{6}t}$

$= \frac{9}{4}pt^2\mathrm{e}^{-\frac{3}{2}t} - 6pt\mathrm{e}^{-\frac{3}{2}t} + 2p\mathrm{e}^{-\frac{3}{2}t} + \frac{1}{36}q\mathrm{e}^{\frac{1}{6}t}$

$36\frac{\mathrm{d}^2T}{\mathrm{d}t^2} + 108\frac{\mathrm{d}T}{\mathrm{d}t} + 81T = 216\mathrm{e}^{-\frac{3}{2}t} + 100\mathrm{e}^{\frac{1}{6}t}$

$\Rightarrow 36\left(\frac{9}{4}pt^2\mathrm{e}^{-\frac{3}{2}t} - 6pt\mathrm{e}^{-\frac{3}{2}t} + 2p\mathrm{e}^{-\frac{3}{2}t} + \frac{1}{36}q\mathrm{e}^{\frac{1}{6}t}\right) + 108\left(-\frac{3}{2}pt^2\mathrm{e}^{-\frac{3}{2}t} + 2pt\mathrm{e}^{-\frac{3}{2}t} + \frac{1}{6}q\mathrm{e}^{\frac{1}{6}t}\right) +$

$81\left(pt^2\mathrm{e}^{-\frac{3}{2}t} + q\mathrm{e}^{\frac{1}{6}t}\right) = 216\mathrm{e}^{-\frac{3}{2}t} + 100\mathrm{e}^{\frac{1}{6}t}$

$\Rightarrow (81p - 162p + 81p)t^2\mathrm{e}^{-\frac{3}{2}t} + (-216p + 216p)t\mathrm{e}^{-\frac{3}{2}t} + (72p)\mathrm{e}^{-\frac{3}{2}t} + (q + 18q +$

$81q)\mathrm{e}^{\frac{1}{6}t} = 216\mathrm{e}^{-\frac{3}{2}t} + 100\mathrm{e}^{\frac{1}{6}t}$

$\Rightarrow (72p)\mathrm{e}^{-\frac{3}{2}t} + (100q)\mathrm{e}^{\frac{1}{6}t} = 216\mathrm{e}^{-\frac{3}{2}t} + 100\mathrm{e}^{\frac{1}{6}t}$

Compare coefficients:

$72p = 216 \Rightarrow p = 3$

$100q = 100 \Rightarrow q = 1$

Particular Integral:

$T = 3t^2\mathrm{e}^{-\frac{3}{2}t} + \mathrm{e}^{\frac{1}{6}t}$

General Solution:

$T = Ate^{-\frac{3}{2}t} + Be^{-\frac{3}{2}t} + 3t^2e^{-\frac{3}{2}t} + e^{\frac{1}{6}t}$

10) b)

Substitute $T = 4$ and $t = 0$

$4 = A \times 0 \times e^0 + Be^0 + 3 \times 0 \times e^0 + e^0$

$\Rightarrow 4 = B + 1$

$\Rightarrow B = 3$

$\therefore T = Ate^{-\frac{3}{2}t} + 3e^{-\frac{3}{2}t} + 3t^2e^{-\frac{3}{2}t} + e^{\frac{1}{6}t}$

$\frac{dT}{dt} = -\frac{3}{2}Ate^{-\frac{3}{2}t} + Ae^{-\frac{3}{2}t} - \frac{9}{2}e^{-\frac{3}{2}t} - \frac{9}{2}t^2e^{-\frac{3}{2}t} + 6te^{-\frac{3}{2}t} + \frac{1}{6}e^{\frac{1}{6}t}$

Substitute $\frac{dT}{dt} = -\frac{1}{3}$ and $t = 0$

$-\frac{1}{3} = -\frac{3}{2}A \times 0 \times e^0 + Ae^0 - \frac{9}{2}e^0 - \frac{9}{2} \times 0 \times e^0 + 6 \times 0 \times e^0 + \frac{1}{6}e^0$

$\Rightarrow -\frac{1}{3} = A - \frac{9}{2} + \frac{1}{6}$

$\Rightarrow A = 4$

$\therefore T = 4te^{-\frac{3}{2}t} + 3e^{-\frac{3}{2}t} + 3t^2e^{-\frac{3}{2}t} + e^{\frac{1}{6}t}$

10) c)

$T = 4te^{-\frac{3}{2}t} + 3e^{-\frac{3}{2}t} + 3t^2e^{-\frac{3}{2}t} + e^{\frac{1}{6}t}$

$\Rightarrow T = (3t^2 + 4t + 3)e^{-\frac{3}{2}t} + e^{\frac{1}{6}t}$

As $t \to \infty$, $e^{-\frac{3}{2}t} \to 0$ and $(3t^2 + 4t + 3)e^{-\frac{3}{2}t} \to 0$ as the exponential term tends to zero faster than the quadratic increases. Also we have $e^{\frac{1}{6}t} \to \infty$ as $t \to \infty$, so for large values of t we have $T \sim e^{\frac{1}{6}t}$

So the temperature increases without bound.

10) d)

$T|_{t=10} = (3 \times 10^2 + 4 \times 10 + 3)e^{-\frac{3}{2} \times 10} + e^{\frac{1}{6} \times 10}$

$= 0.0001049 \ldots + 5.29449 \ldots$

$= 5.29℃$ (3 SF)

10) e)

$20 = (3t^2 + 4t + 3)\mathrm{e}^{-\frac{3}{2}t} + \mathrm{e}^{\frac{1}{6}t}$

By the time the temperature is 20℃, $(3t^2 + 4t + 3)\mathrm{e}^{-\frac{3}{2}t} \approx 0$

So we can estimate the time by solving

$20 = \mathrm{e}^{\frac{1}{6}t}$

$\Rightarrow \frac{1}{6}t = \ln 20$

$\Rightarrow t = 6 \ln 20$

$= 17.97439 \ldots$

$= 18.0$ minutes (3 SF)